JENNIFER L. CROSS

Center for Engineering Education and Outreach
Tufts University
200 Boston Ave, Suite G810
Medford, Massachusetts 02155
jennifer.light.cross@gmail.com
jenncross.com

RESEARCH EXPERTISE

Participatory Design & Development: Collaborative design and rapid prototyping of innovative coding interfaces for robotics education, working directly with users to create intuitive software and tangible interaction paradigms.

Design-Based Research for Human-Robot Interaction: Building empirically-grounded principles of human-robot interaction while simultaneously creating practical educational technologies through systematic research-design cycles

Interdisciplinary Engineering Education: Developing systems and frameworks to create mutually beneficial connections between engineering and other disciplines, enhancing learning and engagement in both domains to support all learners' educational experiences.

ACADEMIC APPOINTMENTS

Research Assistant Professor, Center for Engineering Education and Outreach — 2020 – Present
Part-Time Lecturer, School of Engineering — 2019 – Present
Lead Program Instructor, *Pre-College Engineering Design Lab Program*, University College — 2020 – Present
Tufts University, Medford, MA

Postdoctoral Research Associate — 2018 - 2020
Tufts University, Medford, MA
Center for Engineering Education and Outreach – FET Lab (Future Educational Technologies Lab)
Postdoctoral Advisor: Chris Rogers

Postdoctoral Researcher — 2017 – 2018
Carnegie Mellon University, Pittsburgh, PA
Robotics Institute – CREATE Lab (Community Robotics Education and Technology Empowerment Lab)
Postdoctoral Advisor: Illah Nourbakhsh

EDUCATION

Ph.D. in Robotics — 2017
Carnegie Mellon University, Pittsburgh, PA
Dissertation: Creative Robotic Systems for Talent-Based Learning
Advisor: Illah Nourbakhsh; Committee: Mitchel Resnick (MIT), Jack Mostow (CMU), and Aaron Steinfeld (CMU)

M.S. in Robotics — 2013
Carnegie Mellon University, Pittsburgh, PA

B.S. in Electrical and Computer Engineering — 2010
Franklin W. Olin College of Engineering, Needham, MA
Member of Olin College's fifth graduating class

AWARDS & HONORS

Program for Interdisciplinary Education Research Fellow 2011 – 2017
Institute of Education Sciences – Department of Education

Graduate Research Fellowship Program Fellow 2011 – 2014
National Science Foundation

Best Paper Awards
Honorable Mention - ACM IUI International Conference on Intelligent User Interfaces 2019
Honorable Mention - ACM CHI Conference on Human Factors in Computing Systems 2017
Best Paper - IEEE Integrated STEM Education Conference 2013

Olin College Full Merit Scholarship 2006 – 2010
Franklin W. Olin College of Engineering

PROFESSIONAL EXPERIENCES

The Pennsylvania State University, State College, PA
National Science Foundation EEREU Undergraduate Scholar Summer 2009
Electrical Engineering Department – Radar Space Sciences Lab
Lab Director: John Mathews

General Electric Transportation, Erie, PA
Legacy Locomotive and Reliability Engineering Intern Summer 2008
Locomotive Reliability Group

F. W. Olin College of Engineering, Needham, MA
Undergraduate Research Assistant
Advanced Computing Lab – Self-Directed Exploded Field-Programmable Gate Array Project 2008 – 2010
Lab Director: Mark Chang
Olin Biomimetic Robotics Lab – Self-Directed Robotic Shark Project 2007 – 2008
Lab Director: Gill Pratt

Charles Stark Draper Laboratory, Cambridge, MA
Undergraduate Intern Summer 2007
Mechanical Systems Division and Robotic Systems Division

GRANTS & GIFTS

Awarded Proposals, PI**:**

[1] *CS4ALL: Engaging IEP Teams as Co-Designers of Middle-Grades Computing Activities for Autistic Students* 2023
NSF #2318191
PI: Jennifer Cross, Tufts University

[2] *RETTL: Integrating AI with Smart Engineering and English Language Arts in Upper Elementary Education* 2021
NSF #1932854
PI: Jennifer Cross, Tufts University

[3] *Online Teacher Professional Development - Exploration of Teacher and Classroom Outcomes* 2020
LEGO Foundation
PI: Jennifer Cross, Tufts University

Awarded Proposals, co-PI**:**

[1] *DRK-12: The Smart Playground: Computational Thinking through Robotics in Early Childhood* 2023
NSF # 2301247; 2301248; 2301249
Collaborative Project: University of California Irving; Boston College
PIs: Chris Rogers, Tufts University; Marina Bers, Boston College; Andres Bustamante, UCI

[2] *IUSE: Improving Students' Sociotechnical Literacy in Engineering* 2021
NSF #2110727

PI: Ethan Danahy, Tufts University

[3] *STEM+C: Researching the Integration of Robotics into Middle School Physical Science Courses* 2019
NSF #1932854
PI: Debra Bernstein, TERC

Awarded Co-authored proposals, neither PI nor co-PI**:**

[4] *STTR Phase II: A Low-Cost Robotics Kit for Elementary Education* 2018
NSF #1831177
PI: Tom Lauwers, BirdBrain Technologies

[5] *STTR Phase I: A Low-Cost Robotics Kit for Elementary Education* 2016
NSF #1648747
PI: Tom Lauwers, BirdBrain Technologies

[6] *Flutter Pilot (Elementary Robotics Platform for Systems Engineering and Science)* 2016
Claude Worthington Benedum Foundation
PI: Illah Nourbakhsh, Carnegie Mellon University

[7] *MSP: Creative Robotics: an inclusive program for fostering diverse STEM talent in middle school*) 2013
NSF #1321227
PI: Illah Nourbakhsh, Carnegie Mellon University

PUBLICATIONS

[1] Mutch-Jones, K., Bernstein, D., Cassidy, M., & **Cross, J. L.** (under review). Increasing Teacher Capacity To Enhance Science Instruction With Robot Models And Experiments. Contemporary Issues in Technology and Teacher Education (CITE Journal).

[2] Zhang, Z., Willner-Giwerc, S., Sinapov, J., **Cross, J.,** and Rogers, C. (2022). An Interactive Robot Platform for Introducing Reinforcement Learning to K-12 Students. Robotics in Education (RiE2021). Springer, Cham.

[3] Zito, L., **Cross, J.**, Brewer, B., Speer, S., Tasota, M., Hamner, E., Johnson, M., Lauwers, T. and Nourbakhsh, I. (2021). Leveraging tangible interfaces in primary school math: Pilot testing of the Owlet math program. *International Journal of Child-Computer Interaction*, 27.

[4] Hsu, Y.-C., **Cross, J.**, Dille, P., Tasota, M., Dias, B., Sargent, R., Huang, K., and Nourbakhsh, I. (2020). Smell Pittsburgh: Engaging Community Citizen Science for Air Quality. *ACM Transactions on Interactive Intelligent Systems (TiiS*), 10(4).

[5] Mathews, J. D., Briczinski, S. J., Malhotra, A., and **Cross, J.** (2010). Extensive Meteoroid Fragmentation in V/UHF Radar Meteor Observations at Arecibo Observatory. *Geophysical Research Letters*, 37(4).

CONFERENCE PUBLICATIONS

[1] Burman, T., Dahal, M., Xu, G., Rogers, C., **Cross, J.**, and Sinapov, J. (2025). Smart Motor: A Low-Cost Hardware And Software Toolkit For Introducing Supervised Machine Learning To Elementary School Students. In EAAI'25: Proceedings of the Fifteenth Symposium on Educational Advances in Artificial Intelligence, Philadelphia, PA.

[2] Xu, G., Zabner, D., **Cross, J.**, Nadler, D., Coxon, S., and Engelkenjohn, K. (2023). Conducting the Pilot Study of Integrating AI: An Experience Integrating Machine Learning into Upper Elementary Robotics Learning (Work in Progress). In Proceedings of 2023 American Society for Engineering Education (ASEE) Annual Conference and Exposition. Baltimore, Maryland.

[3] Bernstein, D., Cassidy, M., Mutch-Jones, K., and **Cross, J. L.** (2022). Robots In Science: How Middle School Science Teachers Design Integrated Robotics Units For Their Science Classes. *In Proceedings of the 16th International Conference of the Learning Sciences (ICLS)*, pp. 1968-1969. International Society of the Learning Sciences.

[4] **Cross, J.,** Brewer, B., Hamner, E., Zito, L., Speer, S., and Tasota, M. (2020). Pilot Results of a Digital Manipulative for Elementary Mathematics. Paper presented at the *2020 Annual Meeting of the American Educational Research Association (AERA)*, San Francisco, CA.

[5] Hsu, Y.-C., **Cross, J.**, Dille, P., Tasota, M., Dias, B., Sargent, R., Huang, K., and Nourbakhsh, I. (2019). Smell Pittsburgh: Community-Empowered Mobile Smell Reporting System. In *Proceedings of the 24th International Conference on Intelligent User Interfaces (ACM IUI '19)*, Marina del Ray, CA.

[6] Hsu, Y.-C., **Cross, J.**, Dille, P., Nourbakhsh, I., Leiter, L., and Grode, R. (2018). Visualization Tool for Environmental Sensing and Public Health Data. *In Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems (DIS '18 Companion),* Hong Kong.

[7] **Cross, J.**, Hamner, E., Zito, L., and Nourbakhsh, I. (2017). Student Outcomes from the Evaluation of a Transdisciplinary Middle School Robotics Program. *In Proceedings of 2017 IEEE Frontiers in Education Conference (FIE)*, Indianapolis, Indiana.

[8] Hamner, E., Zito, L., **Cross, J.**, Tasota, M., Dille, P., Fulton, S., Johnson, M., Nourbakhsh, I., and Schapiro, J. (2017). Development and Results from User Testing of a Novel Robotics Kit Supporting Systems Engineering for Elementary-Aged Students. *In Proceedings of 2017 IEEE Frontiers in Education Conference (FIE)*, Indianapolis, Indiana.

[9] Hsu, Y.-C., Dille, P., **Cross, J.**, Dias, B., Sargent, R., and Nourbakhsh, I. (2017). Community-Empowered Air Quality Monitoring System. *In Proceedings of 2017 ACM CHI Conference on Human Factors in Computing Systems*, Denver, Colorado**. (Honorable Mention Award)**

[10] **Cross, J.**, Hamner, E., Zito, L., Nourbakhsh, I., and Bernstein, D. (2016). Development of an Assessment for Measuring Middle School Student Attitudes towards Robotics Activities. *In Proceedings of 2016 IEEE Frontiers in Education Conference (FIE)*, Erie, Pennsylvania.

[11] **Cross, J.**, Hamner, E., Zito, L., and Nourbakhsh, I. (2016). Engineering and Computational Thinking Talent in Middle School Students: a Framework for Defining and Recognizing Student Affinities. *In Proceedings of 2016 IEEE Frontiers in Education Conference (FIE)*, Erie, Pennsylvania.

[12] Hamner, E., Zito, L., **Cross, J.**, Slezak, B., Mellon, S., Harapko, H., and Welter, M. (2016). Utilizing Engineering to Teach Non-Technical Disciplines: Case Studies of Robotics within Middle School English and Health Classes. *In Proceedings of 2016 IEEE Frontiers in Education Conference (FIE)*, Erie, Pennsylvania.

[13] Hamner, E., **Cross, J.**, Zito, L., Bernstein, D., and Mutch-Jones, K. (2016). Training Teachers to Integrate Engineering into Non- Technical Middle School Curriculum. *In Proceedings of 2016 IEEE Frontiers in Education Conference (FIE)*, Erie, Pennsylvania.

[14] Bernstein, D., Mutch-Jones, K., Hamner, E., and **Cross, J.** (2015). Robots and Romeo and Juliet: Studying Teacher Integration of Robotics into Middle School Curricula. Paper presented at the 2016 Annual Meeting of the American Educational Research Association (AERA), Washington, DC.

[15] **Cross, J.**, Hamner, E., Bartley, C., and Nourbakhsh, I. (2015). Arts & Bots: Application and Outcomes of a Secondary School Robotics Program. *In Proceedings of 2015 IEEE Frontiers in Education Conference (FIE)*, El Paso, Texas.

[16] **Cross, J.** and Hamner, E. (2014). Identifying and Cultivating Diverse STEM Talent through Creative Robotics. *In Proceedings of 2014 American Society for Engineering Education (ASEE) Annual Conference and Exposition*, Indianapolis, Indiana.

[17] **Cross, J.**, Bartley, C., Hamner, E., and Nourbakhsh, I. (2013). A Visual Robot-Programming Environment for Multidisciplinary Education. *In Proceedings of 2013 IEEE International Conference on Robotics and Automation (ICRA*), Karlsruhe, Germany.

[18] Hamner, E. and **Cross, J.** (2013). Arts & Bots: Techniques for distributing a STEAM robotics program through K-12 classrooms. *In Proceedings of the 2013 IEEE Integrated STEM Education Conference (ISEC)*, Princeton, NJ. **(Best Paper Award)**

[19] Brown, H. B., Nourbakhsh, I., Bartley, C., **Cross, J.**, Dille, P., Schapiro, J., and Styler, A. (2012). ChargeCar Community Conversions: Practical, Electric Commuter Vehicles Now! *In Proceedings of the 2012 IEEE International Electric Vehicle Conference (IEVC)*, Greenville, SC.

TEACHING

Introduction to Computing in Engineering 2019 - 2025
Course Instructor, Tufts University
Audience: 40 first-year engineering students annually
Teaches foundational programming and computational methods for engineering problem-solving using Python. Students develop technical skills in algorithmic thinking, data analysis, visualization, and modeling of engineering systems through hands-on projects spanning multiple engineering disciplines. Adapted for remote learning in 2021.

Engineering Design Lab – Pre-College Intensive 2020 -2024
Course Instructor, Tufts University
Audience: 45 high school students, two sessions annually
Introduces Python, robotics concepts (motion, sensing, machine learning, computer vision), and hardware prototyping skills (CAD, laser cutting, 3D printing, woodworking, electronic circuits) through scaffolded challenges and team projects. Adapted for remote learning in 2020 and 2021.

Human-Robot Interaction 2019
Course Instructor, Tufts University
Audience: 29 graduate and senior undergraduate computer science and cognitive science students
Students applied principles of human-robot interaction, experimental design, and user testing methodologies through team-based projects. By course completion, students designed and conducted original human-robot interaction studies that integrated technical implementation with empirical research.

Engineering and Science for Middle and High School Educators I 2018
Course Instructor, Tufts University
Audience: 25 professionals in an engineering education certificate program
An asynchronous online course covering the fundamentals of robotic systems, feedback, and PID control through activities and projects using LEGO® robotics platforms and culminating in student-selected final projects.

Human-Machine System Design 2018
Assistant Instructor with J. Intriligator, Tufts University
Audience: 33 senior and M.S. students in engineering psychology, human factors, or mechanical engineering

Mobile Robotics Project Course (Summer Academy for Math and Science) 2014
Course Instructor, Carnegie Mellon University
Audience: Precollege students in CMU's summer bridge program preparing for STEM majors

Systems Engineering 2012
Teaching Assistant for M. Bergerman, Carnegie Mellon University
Audience: Master's students in a robotic system design, 2-year professional degree program

GUEST LECTURES

Human Factors Engineering Seminar, Tufts University 2023
Topic: Participatory Design of Educational Robots and Technologies with Teachers

Human Robot Interaction Seminar, Tufts University 2023
Topic: Participatory Design of Educational Robots and Technologies with Teachers
Topic: Integrating Instructional Technology

Methods & Materials for Elementary Teachers, West Liberty University, West Liberty, WV 2015
Topic: Transdisciplinary Integration of Creative Robotics

OUTREACH & SERVICE

Tufts University CEEO Diversity Committee 2021 – 2023

Introduction to Robotics for Classrooms 2011 – 2022
Lead Workshop Instructor, Online and various locations including PA; WV; IN; Bristol, UK; and others

Audience: K-12 Educators
Over 200 teachers have participated in my one- and two-day professional development programs

CONTEXT: Tech and Data Fluency for Teaching and Learning Conference 2015 & 2017
Session Leader, Pittsburgh, PA
Topic: Recognizing Student Engineering and Computational Thinking Talents in Transdisciplinary Projects
Audience: Educational technology conference for school administrators and K-12 educators

Integrating the E in STEM Workshop Series 2016
Workshop Leader, Erie, PA
Topic: Transdisciplinary Integration of Creative Robotics for Identification of Student STEM Affinities
Audience: K-12 Educators

OurCS: Opportunities for Undergraduate Research in Computer Science 2013 & 2015
Graduate Organizer, Carnegie Mellon University
Audience: Women in Undergraduate Computer Science Programs
Three-day hackathon-style research workshop for undergraduate computer science students

Robotics Institute Ph.D. Admissions Committee 2012 – 2014
Committee Member

Women@SCS Creative Technology Nights 2012 – 2014
Workshop Leader, Carnegie Mellon University
Topic: Robot Programming with Scratch
Audience: Girls, 11 to 14 years old

Women@SCS Computer Science Roadshows 2011 – 2013
Graduate Student Presenter, Carnegie Mellon University
Audience: K-12 Students and Educators

Peer Review
- *Associate Editor*, IEEE International Conference on Robotics and Automation (ICRA)
- *Review Panelist,* National Science Foundation
- *External Thesis Reviewer*, Aalto University, Finland
- *Reviewer*, Journal of Engineering Education (JEE)
- *Reviewer*, IEEE Robotics and Automation Letters (RA-L)
- *Reviewer*, IEEE Frontiers in Education Conference (FIE)
- *Reviewer*, IEEE Robotics and Automation Magazine (RA-M)

MENTORING

Ph.D. Qualifier Committees
Carnegie Mellon University*: Y.-C. Hsu, E. Avrunin*

Master's Thesis Committees
Tufts University*: M. Bennie*
Carnegie Mellon University: *X. Zhang, M. Bernstein*

Other Graduate and Research Mentorship
Tufts University: *T. Burman, Z. Zhang, E. Goroza, K. Battel*
Boston College: *J. Blake-West*
Carnegie Mellon University*: S. Speer*, *L. Zito, M. Tasota*

Undergraduate Mentorship
Center for Engineering Education and Outreach, Summer Interns Program, Tufts University 2019-2024
Computer Science, Electrical and Computer Engineering Capstone Project Advisor, Tufts University 2018-2019